Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/10/2016
PI Name:	Main, Bob M.S.		
Project Title:	Development and Recommendation of Options for Variable Focus Eyewear for Spaceflight and Mobile Device Technology for In-flight Vision Testing		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
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Space Biology Special Category:	None		
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Comments:			
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	Our project objectives have been to find, investigate, and recommend to NASA, vision related technologies/companies that are developing technology with two main focuses; 1) variable power eyeglasses (manual and electric) that have the ability to change the power of the lenses inflight as needed and 2) miniaturized vision testing equipment that can help NASA monitor and diagnose vision problems as they occur inflight. To accomplish these objectives, we have: Conducted a global search for all related technologies. We did this by accessing our global network of industry contacts and attended technology related conferences. We fully investigated a significant number of companies that have potential technology to help NASA solve several vision related issues. In many cases, we were able to personally see the technology in action (or test it in our office), and met with their technology team/executives (several at their offices) to fully evaluate if the company can deliver the technology (most are in early stages of development). In the past year I personally traveled over 41k miles to find and fully investigate technologies/companies to insure that all relevant		

Task Description:	technologies have been identified and vetted. We also conducted astronaut user panels to get real-space-usability feedback about potential technology. Specifically relevant to each objective: • Adjustable Power Eyeglasses – Investigated 12 companies/technologies and visited/met with eight of these companies and their senior management. I recommended two of these technologies to Dr. Gibson/NASA. (One is currently on the International Space Station (ISS) and the other is being flight certified). • Miniaturized Vision Testing Equipment – Investigated 19 companies/technologies and met with five of the companies' executives or got an in-person demo. Additionally, four of the companies sent their device to me for evaluation. Three of these technologies I recommended to Dr Gibson/NASA (one is currently in clinical trial at the NASA Vision Clinic for potential use on the ISS). The results/findings have been: • Recommended three technologies to NASA that are in different stages of evaluation in the NASA Vision Clinic or are currently on the ISS. • Prepared a detailed set of specifications for the variable power eyeglasses project. NASA and the NSBRI (National Space Biomedical Research Institute) can use these specifications as a baseline for the short and long term process of developing variable power eyeglasses in partnership with several companies. • Discovered, investigated (onsite in many cases), and fostered close relationships with many eye care technology companies that have the ability to deliver current and future technology to help NASA. • Investigated and eliminated several companies that have technology that NASA asked us to evaluate and deemed they were not useful for NASA. • Discovered, recommended, and assisted two companies that applied for the SMARTCAP program (one of these companies, e-Vision, was funded).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It was not the intent of the project to invent new technology but to discover, investigate, and work with companies that already have technologies or are developing technologies that will help NASA with the vision issues astronauts are having. However, the technologies/companies we have investigated (and several recommended to NASA now on the ISS) will also have many benefits for the commercial market. While many of the technologies we have investigated (and continue to monitor) are still in early stages of development, several are either currently available or are soon to be released into the market. We are working with all of these companies to make sure the necessary modifications are made so their technology can be used to help the space program along with their Earth commercialization benefits.
Task Progress:	Since receiving the NSBRI SMARTCAP grant in Feb 2015, we have been able to find, research, and investigate many vision related technologies globally that can help NASA with the vision changes astronauts are experiencing due to the hyperopic-shift issue on long duration flights. We have recommended three of these technologies to NASA and they are in various stages of being evaluated at the NASA Vision Clinic and/or flight certified to be used on the ISS. One technology is currently on the ISS. We are also working closely with several companies globally to develop electronic adjustable power eyeglasses and a handheld retinal imaging device. During the grant period, it became clear that, in order to accomplish the deliverables, there was a need to change the approach on how the companies/technologies would be researched and investigated. Additionally, as new companies/technologies were discovered during the grant period, there was a need to do an on-site visit of the companies to insure that my recommended companies/technologies to NASA were accurate, had good quality control, and were viable. Therefore, a significant amount of additional time and travel was necessary than originally projected in the grant proposal. After conferring with NSBRI, the funds that were earmarked to invest in some technologies were diverted to the additional time and travel part of the budget. In many respects, the funds were invested in the companies (and they received value) as I was able, after vetting the company, to connect them directly with NASA and get their technologies either on the ISS or in process to get on the ISS, which has great PR value. These changes in budget will be reflected in the final budget report and a comprehensive travel log will be submitted.
Bibliography Type:	Description: (Last Updated:)